The vice versa, though, doesn't hold, unfortunately!

An example will help understand why: suppose we have a simple Bloom filter with 4 bits and 2 hash functions. Initially our buffer is empty:

```
B = [0, 0, 0, 0]
```

First, we insert the value 1 into the Bloom filter. Assume we had chosen our hash functions such that $h_0(1) = 0$ and $h_1(1) = 2$, so the key 1 maps to indices {0, 2}; after updating it, our buffer now looks like this:

```
B = [1, 0, 1, 0]
```

Now we insert the value 2, and turns out that $h_0(2) = 1$ and $h_1(2) = 2$. Therefore, now we have:

```
B = [1, 1, 1, 0]
```

Finally, suppose that $h_0(3) = 1$ and $h_1(3) = 0$: if we lookup for the value 3 after those two insertions, both bits at indices 1 and 0 will have been set to 1 even if we never stored 3 in our instance of the filter! Therefore, 3 would give us a false positive.

On the other hand, if we had a different mapping from the hash functions, for instance $h_0(3) = 3$ and $h_1(3) = 0$, then,